Ratios, Rates & Data

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Ratios and proportions**

Simplify the ratio $18 : 24$ to lowest terms.

Divide both terms by 6: $3 : 4$.

2 **[Foundational]** • **Target: Ratios and proportions**

Solve the proportion $\frac{5}{8} = \frac{x}{40}$.

Cross-multiply: $8x = 200$, so $x = 25$.

3 **[Foundational]** • **Target: Ratios and proportions**

A recipe requires flour and sugar in the ratio $3 : 2$. If 12 cups of flour are used, how much sugar is needed?

Sugar $= \frac{2}{3} \times 12 = 8$ cups.

4 **[Foundational]** • **Target: Ratios and proportions**

If $a : b = 4 : 7$ and $b = 35$, find a.

$\frac{a}{4} = \frac{35}{7} = 5$, so $a = 20$.

5 **[Foundational]** • **Target: Unit rates**

A car uses 8 gallons of gas to travel 224 miles. Find its fuel efficiency in miles per gallon.

$\frac{224}{8} = 28$ miles per gallon.

6 **[Foundational]** • **Target: Unit rates**

A printer produces 150 pages in 6 minutes. Find its rate in pages per minute, and use it to find how long it takes to print 400 pages.

Rate $= \frac{150}{6} = 25$ pages per minute. Time $= \frac{400}{25} = 16$ minutes.

7 **[Foundational]** • **Target: Percentage change and applications**

A price increases from 60 to 75 dollars. Find the percentage increase.

$\frac{75 - 60}{60} \times 100 = 25\%$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Percentage change and applications

A jacket originally 120 dollars is discounted 30 percent. Find the sale price.

$120 \times 0.7 = 84$ dollars.

9 [Intermediate] • Target: Percentage change and applications

A population decreases by 15 percent one year, then increases by 15 percent the next. Is the final population the same as the original?

Start with 100. After a 15\% decrease: 85. After a 15\% increase: $85 \times 1.15 = 97.75$. No — the final population is 2.25\% less than the original.

10 [Intermediate] • Target: Percentage change and applications

An investment of 2000 dollars grows to 2300 dollars after one year. Find the percentage increase.

$\frac{2300 - 2000}{2000} \times 100 = 15\%$.

11 [Intermediate] • Target: Weighted averages

A student scores 80 on a test worth 40 percent of the grade and 90 on a test worth 60 percent of the grade. Find the weighted average grade.

$0.4(80) + 0.6(90) = 32 + 54 = 86$.

12 [Intermediate] • Target: Weighted averages

A store sells 3 items at 10 dollars each and 7 items at 20 dollars each. Find the average price per item sold.

Total revenue $= 3(10) + 7(20) = 30 + 140 = 170$ dollars. Total items $= 10$. Average $= \frac{170}{10} = 17$ dollars.

13 [Intermediate] • Target: Scale and similar figures

A model car is built at a scale of $1 : 24$. If the actual car is 4.8 meters (480 cm) long, find the length of the model, in centimeters.

$\frac{480}{24} = 20$ centimeters.

14 [Intermediate] • Target: Scale and similar figures

A map has a scale of 1 cm to 5 km. If two cities are 8.5 cm apart on the map, find the actual distance between them.

$8.5 \times 5 = 42.5$ km.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Interpreting data tables** • **Cross-Lesson: 5.01 Statistics & Probability**

A table shows the hours worked by 4 employees in a week: A: 32, B: 40, C: 28, D: 36. Find the total hours worked and the mean hours per employee.

Total $= 32 + 40 + 28 + 36 = 136$. Mean $= \frac{136}{4} = 34$ hours.

16 **[Challenging]** • **Target: Interpreting data tables** • **Cross-Lesson: 6.01 Geometry**

A table shows quarterly revenue, in thousands of dollars: Q1: 50, Q2: 65, Q3: 70, Q4: 55. Find the percentage of annual revenue earned in Q3, to the nearest tenth of a percent.

Total $= 50 + 65 + 70 + 55 = 240$. Q3 percentage $= \frac{70}{240} \times 100 \approx 29.2\%$.

17 **[Challenging]** • **Target: Bar graphs and data displays** • **Cross-Lesson: 7.01 Trigonometry & Additional Topics**

The bar graph shows a company's monthly sales, in thousands of dollars. Find the percentage increase in sales from January to March.

$\frac{65 - 40}{40} \times 100 = 62.5\%$.

18 **[Challenging]** • **Target: Proportional reasoning in context** • **Cross-Lesson: 8.01 Exponentials & Logarithms**

A factory produces 450 units using 6 machines running for 8 hours. Assuming the same rate per machine-hour, how many units would 9 machines produce in 8 hours?

Rate $= \frac{450}{6 \times 8} = \frac{450}{48} = 9.375$ units per machine-hour. New machine-hours $= 9 \times 8 = 72$. Units $= 72 \times 9.375 = 675$.

19 **[Challenging]** • **Target: Proportional reasoning in context** • **Cross-Lesson: 1.01 Heart of Algebra**

If 5 workers can build a fence in 12 days, how many days would it take 4 workers, assuming the same work rate per worker?

Total work $= 5 \times 12 = 60$ worker-days. Days for 4 workers $= \frac{60}{4} = 15$ days.

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20 [Challenging] • Target: A well-designed word problem: customer growth for a small business • Cross-Lesson: 2.01 Quadratics & Polynomials

This question has three parts. In the first quarter, a business had 240 customers, and 180 of them made a repeat purchase. (a) Find the repeat-purchase rate as a percentage. (b) In the second quarter, the number of customers grew by 25 percent, and the repeat-purchase rate held steady at the percentage found in part (a). Find the number of repeat customers in the second quarter. (c) The business sets a goal to raise the repeat-purchase rate to 80 percent by the third quarter, while keeping the same total number of customers as the second quarter. Find how many additional repeat customers (beyond part (b)'s total) are needed to reach this goal.

(a) $\frac{180}{240} \times 100 = 75\%$. (b) New customers $= 240 \times 1.25 = 300$. Repeat customers $= 0.75 \times 300 = 225$. (c) 80\% of 300 is $0.8 \times 300 = 240$ repeat customers needed. Additional customers needed $= 240 - 225 = 15$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Ratios and proportions

Solve the proportion $\frac{14}{4} = \frac{x}{34}$ for x.

a. 121

b. 119 ✓

c. 117

d. 4

Cross-multiply: $4x = 14 \times 34 = 476$, so $x = 119$.

22 [Foundational] • Target: Unit rates

Solve the proportion $\frac{4}{5} = \frac{x}{30}$ for x.

a. 26

b. 22

c. 24 ✓

d. 5

Cross-multiply: $5x = 4 \times 30 = 120$, so $x = 24$.

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23 [Foundational] • Target: Percentage change and applications

Solve the proportion $\frac{3}{7} = \frac{x}{7}$ for x.

a. 1

b. 3 ✓

c. 7

d. 5

Cross-multiply: $7x = 3 \times 7 = 21$, so $x = 3$.

24 [Foundational] • Target: Weighted averages

Solve the proportion $\frac{10}{6} = \frac{x}{42}$ for x.

a. 70 ✓

b. 6

c. 72

d. 68

Cross-multiply: $6x = 10 \times 42 = 420$, so $x = 70$.

25 [Foundational] • Target: Scale and similar figures

Solve the proportion $\frac{2}{8} = \frac{x}{28}$ for x.

a. 8

b. 9

c. 5

d. 7 ✓

Cross-multiply: $8x = 2 \times 28 = 56$, so $x = 7$.

26 [Foundational] • Target: Interpreting data tables

Solve the proportion $\frac{8}{8} = \frac{x}{29}$ for x.

a. 31

b. 8

c. 29 ✓

d. 27

Cross-multiply: $8x = 8 \times 29 = 232$, so $x = 29$.

27 [Foundational] • Target: Bar graphs and data displays

Solve the proportion $\frac{12}{3} = \frac{x}{46}$ for x.

a. 184 ✓

b. 186

c. 182

d. 3

Cross-multiply: $3x = 12 \times 46 = 552$, so $x = 184$.

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28 [Foundational] • Target: Proportional reasoning in context

A budget decreases from 40 to 20. Find the percentage decrease.

a. 50% ✓

b. 55%

c. 100%

d. 45%

$\frac{20}{40} \times 100 = 50\%$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A well-designed word problem: customer growth for a small business

An attendance figure increases from 240 to 312. Find the percentage increase.

a. 30% ✓

b. 25%

c. 35%

d. 60%

$\frac{72}{240} \times 100 = 30\%$.

30 [Intermediate] • Target: Ratios & Proportional Reasoning

A membership total increases from 60 to 78. Find the percentage increase.

a. 30% ✓

b. 35%

c. 25%

d. 60%

$\frac{18}{60} \times 100 = 30\%$.

31 [Intermediate] • Target: Percent Changes & Financial Calculations

A membership total decreases from 60 to 30. Find the percentage decrease.

a. 55%

b. 100%

c. 45%

d. 50% ✓

$\frac{30}{60} \times 100 = 50\%$.

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32 [Intermediate] • Target: Speed, Distance & Average Rates

An attendance figure increases from 160 to 232. Find the percentage increase.

a. 40%

b. 50%

c. 45% ✓

d. 90%

$\frac{72}{160} \times 100 = 45\%$.

33 [Intermediate] • Target: Data Tables & Two-Way Frequency

A price increases from 260 to 377. Find the percentage increase.

a. 40%

b. 90%

c. 45% ✓

d. 50%

$\frac{117}{260} \times 100 = 45\%$.

34 [Intermediate] • Target: Ratios and proportions

An attendance figure decreases from 60 to 51. Find the percentage decrease.

a. 10%

b. 20%

c. 30%

d. 15% ✓

$\frac{9}{60} \times 100 = 15\%$.

35 [Intermediate] • Target: Unit rates

A student scores 85 on an assignment worth 20% of the grade and 50 on an assignment worth 80% of the grade. Find the weighted average grade.

a. 60

b. 67

c. 57 ✓

d. 54

$0.2(85) + 0.8(50) = 17 + 40 = 57$.

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36 [Intermediate] • Target: Percentage change and applications

A student scores 65 on an assignment worth 40% of the grade and 95 on an assignment worth 60% of the grade. Find the weighted average grade.

a. 87 **b.** 80

c. 83 ✓ **d.** 86

$0.4(65) + 0.6(95) = 26 + 57 = 83$.

37 [Intermediate] • Target: Weighted averages

If 6 workers can complete a task in 11 days, how many days would it take 3 workers, assuming the same work rate per worker?

a. 11 **b.** 20

c. 22 ✓ **d.** 24

Total work $= 6 \times 11 = 66$ worker-days. Days for 3 workers $= \frac{66}{3} = 22$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Scale and similar figures • Cross-Lesson: 8.01 Exponentials & Logarithms

If 9 workers can complete a task in 12 days, how many days would it take 6 workers, assuming the same work rate per worker?

a. 16 **b.** 12

c. 20 **d.** 18 ✓

Total work $= 9 \times 12 = 108$ worker-days. Days for 6 workers $= \frac{108}{6} = 18$.

39 [Challenging] • Target: Interpreting data tables • Cross-Lesson: 1.01 Heart of Algebra

A map has a scale of 1 cm to 17 km. If two cities are 15.5 cm apart on the map, find the actual distance between them.

a. 267.5 **b.** 265.5

c. 261.5 **d.** 263.5 ✓

$15.5 \times 17 = 263.5$ km.

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40 [Challenging] • Target: Bar graphs and data displays • Cross-Lesson: 2.01 Quadratics & Polynomials

A map has a scale of 1 cm to 11 km. If two cities are 6.5 cm apart on the map, find the actual distance between them.

a. 69.5

b. 71.5 ✓

c. 73.5

d. 75.5

$6.5 \times 11 = 71.5$ km.

41 [Challenging] • Target: Ratios & Proportional Reasoning • Cross-Lesson: 3.01 Functions & Graphs

The ratio of red, blue, and yellow marbles in a bag is $3 : 4 : 5$. If there are 48 marbles in total, how many blue marbles are there?

a. 12

b. 16 ✓

c. 20

d. 24

Total ratio parts $= 3 + 4 + 5 = 12$. Each part $= \frac{48}{12} = 4$. Blue marbles $= 4 \times 4 = 16$.

42 [Challenging] • Target: Ratios & Proportional Reasoning • Cross-Lesson: 5.01 Statistics & Probability

A map has a scale of 1 cm $: 25$ km. If two cities are 7.2 cm apart on the map, what is the actual distance between them?

a. 150 km

b. 175 km

c. 180 **km** ✓

d. 200 km

Actual distance $= 7.2 \times 25 = 180$ km.

43 [Challenging] • Target: Percent Changes & Financial Calculations • Cross-Lesson: 6.01 Geometry

If 40% of x is equal to 30% of y, what is the ratio $\frac{x}{y}$?

a. $\frac{3}{4}$ ✓

b. $\frac{4}{3}$

c. $\frac{3}{7}$

d. $\frac{7}{3}$

$0.40x = 0.30y \Rightarrow \frac{x}{y} = \frac{0.30}{0.40} = \frac{3}{4}$.

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44 [Challenging] • Target: Speed, Distance & Average Rates • Cross-Lesson: 7.01 Trigonometry & Additional Topics

Train A leaves a station traveling east at 80 km/h. Two hours later, Train B leaves the same station traveling east on a parallel track at 120 km/h. How many hours after Train B departs will it overtake Train A?

a. 3

b. 4 ✓

c. 5

d. 6

Head start of Train A $= 80 \times 2 = 160$ km. Relative closing speed $= 120 - 80 = 40$ km/h. Time $= \frac{160}{40} = 4$ hours.

45 [Challenging] • Target: Data Tables & Two-Way Frequency • Cross-Lesson: 8.01 Exponentials & Logarithms

In a survey of 200 high school seniors, 120 plan to attend college, 50 plan to work, and 30 are undecided. If a student is chosen at random, what is the probability that the student does NOT plan to work?

a. $\frac{1}{4}$

b. $\frac{3}{4}$ ✓

c. $\frac{3}{5}$

d. $\frac{17}{20}$

Students not planning to work $= 200 - 50 = 150$. Probability $= \frac{150}{200} = \frac{3}{4}$.

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